

Yasuda Lab

User Manual of FLIMage

Two Photon Fluorescence Lifetime Imaging Microscope Software

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FLIMage! Window Overview

FLIMage! Main window

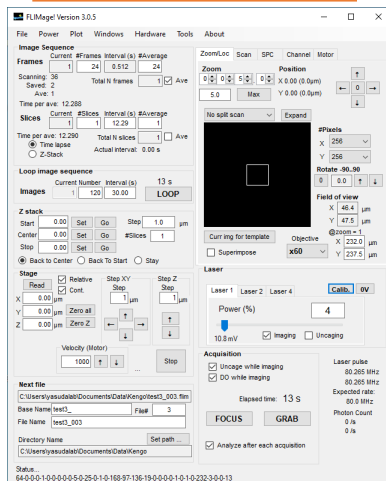
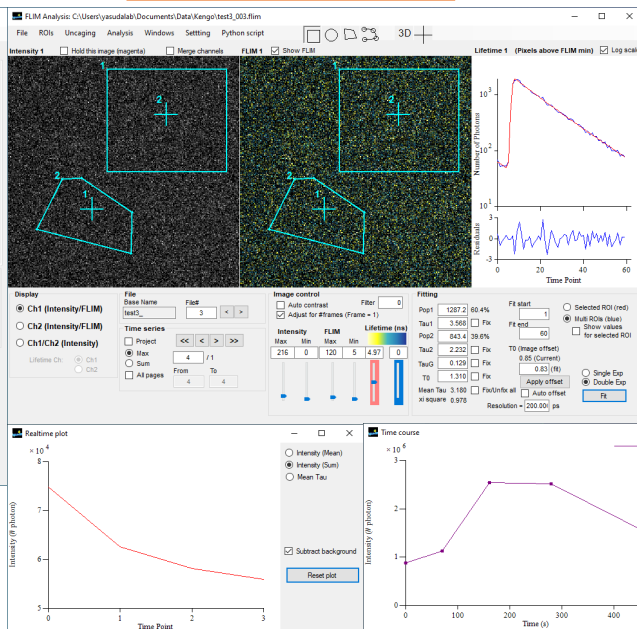
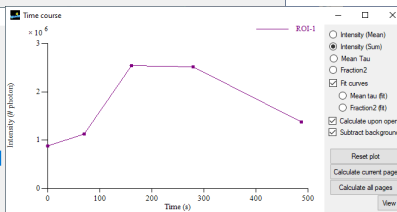


Image display window

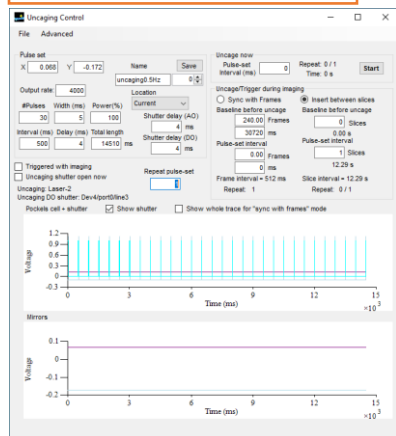


Realtime plot window

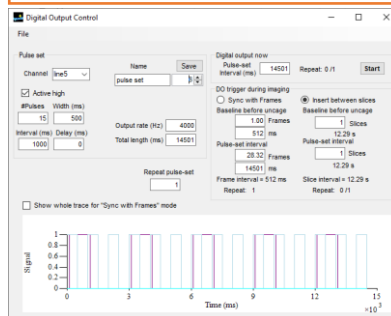
Time course window



Uncaging control window



Digital Output control window



FLIMage! Main Window

This is the main window to control FLIM image acquisition.

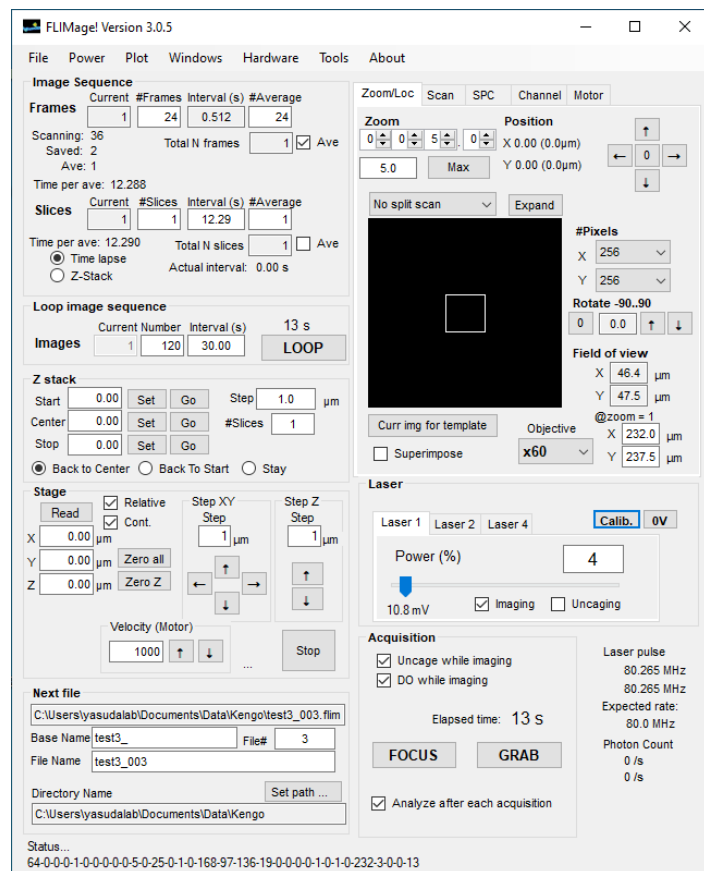


Image sequence

Frames

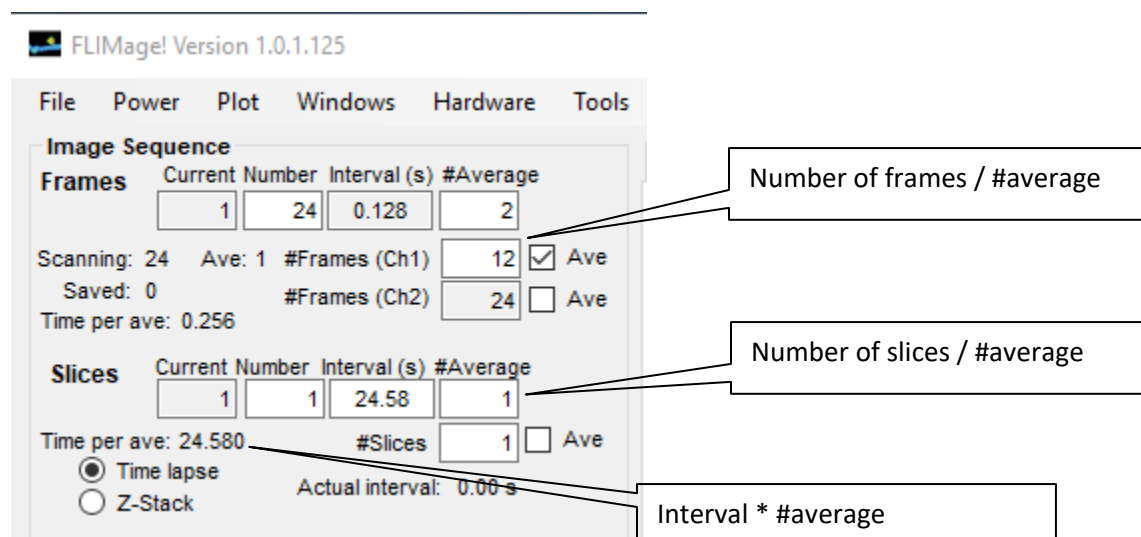
Frames are acquired continuously in a single sequence with an interval defined by the scanning speed. For example, if scanning speed is 2 ms per line and 128 lines per frame, frames are acquired every 256 ms. Shutter does not close in between frames.

You can average frames to increase the signal to noise ratio (Put a value > 1 in #Average box and check “Ave” check box). If averaging on/off is different for different channels, images from different channels are saved in different file.

The textbox below “Ave” checkbox indicates the number of averaged frame (Number of frames / #average). You can edit either the averaged frame or total number of frames, and the other value will be automatically changed.

Current frame displays the progress of display, which maybe delayed from actual acquisition.

“Saved frames/slices” displays the progress of file saving.



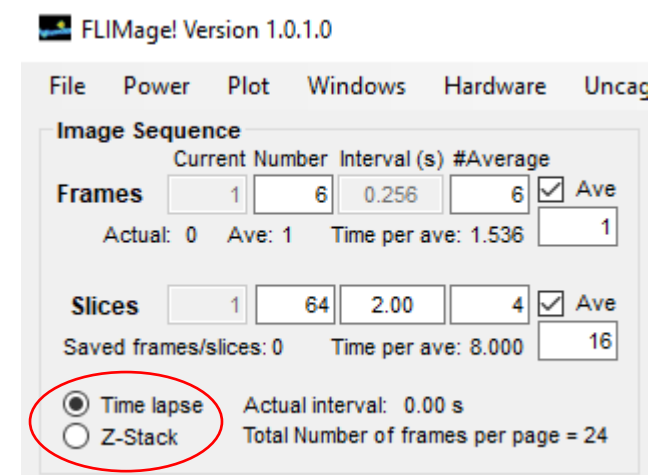
Slices

Slices are a set of frames (see above). Image acquisition is stopped after each “frames” acquisition. You can set an interval time, but the interval has to be longer than the total “frame” acquisition (if you set shorter time, it is automatically updated to a longer time). You can also average slices.

You can average multiple slices in the “Time lapse” mode. To do so, put a value > 1 in #Average box and check “Ave” check box.

All frames and slices are saved in one file.

If you choose “Z-Stack”, motor will move one step (step size defined in “Z-stack” panel) between slices. Again, only 1 averaged frame is allowed per 1 slice in Z-stack acquisition.

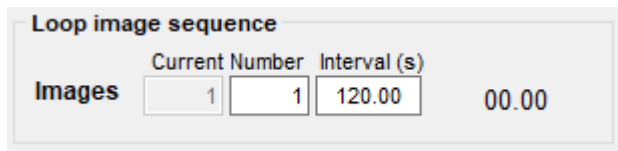


Time lapse / Z-stack

This will change the mode of image acquisition. See above.

Loop image sequences

You can loop the above image sequence. Set number of images you want to acquire and intervals of the sequence. Images are saved in different files.

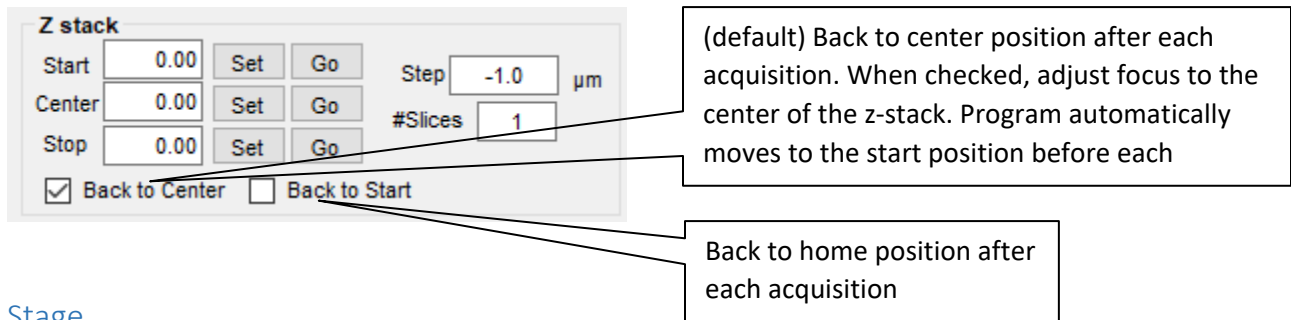


Loop image sequence

	Current Number	Interval (s)
Images	1	120.00

Z-stack

This panel allows you to set the number of slices and steps for Z stack.



Z stack

Start	0.00	Set	Go	Step	-1.0	μm
Center	0.00	Set	Go	#Slices	1	
Stop	0.00	Set	Go			

☒ Back to Center ☐ Back to Start

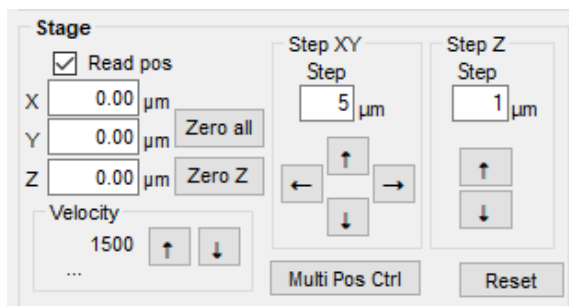
(default) Back to center position after each acquisition. When checked, adjust focus to the center of the z-stack. Program automatically moves to the start position before each

Back to home position after each acquisition

Stage

This controls MP-285 motor stage. It continuously read the position. To change the speed of the motion, press arrow keys in "velocity" panel.

Reset button will try to recover the communication. Often you have to restart MP-285 before pressing this.



Stage

☒ Read pos

X 0.00 μm Y 0.00 μm Z 0.00 μm Zero all Zero Z

Velocity 1500 ↑ ↓

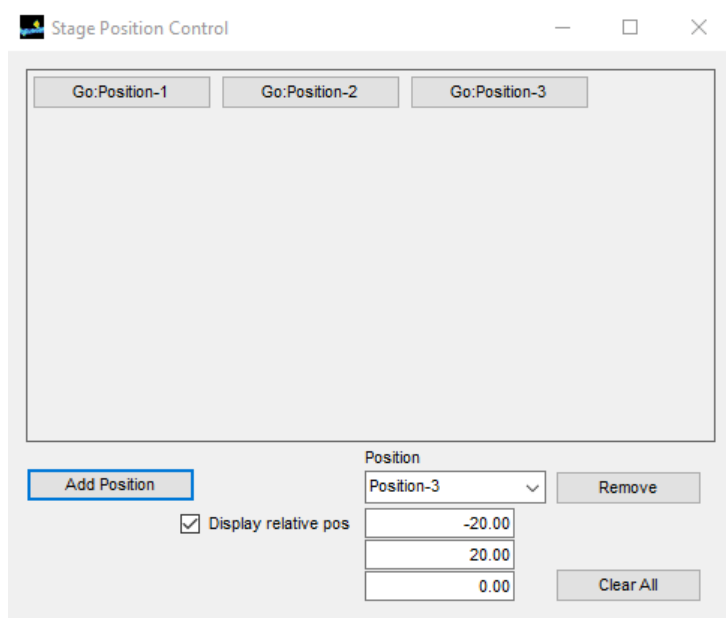
Step XY Step 5 μm ↑ ↓ ← →

Step Z Step 1 μm ↑ ↓

Multi Pos Ctrl Reset

Managing multiple motor position

By clicking the “Stage Control” menu in the “Tools” panel, you can manage multiple stage position using “Stage Position Control” window (see below). By clicking “Add position”, it will create a button (“Go Position-1”). You can create as many buttons as you want, and then go back to the position by clicking the button. To remove specific position, use “Position” pulldown to select one, and press remove.



Next file

The file name will be defined as:

(Base Name) (3 digit File number).flim

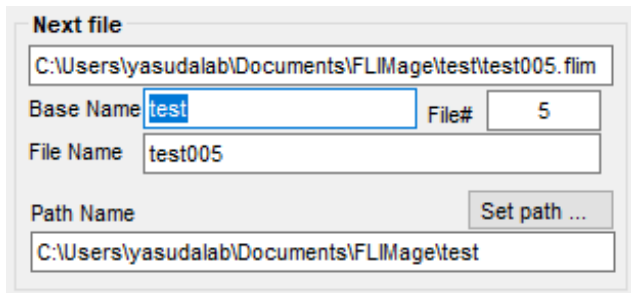
For example, if you put “Experimetn_1A” as the base name, the first imaged will be save in:

Experimetn_1A001.flim

The file number will automatically increase by 1 after finishing each image sequence. The next image will be saved in:

Experimetn_1A002.flim

You need to set the save path by pressing “save path...” button.



Next file

C:\Users\yasudalab\Documents\FLImage\test\test005.flim

Base Name File#

File Name

Path Name

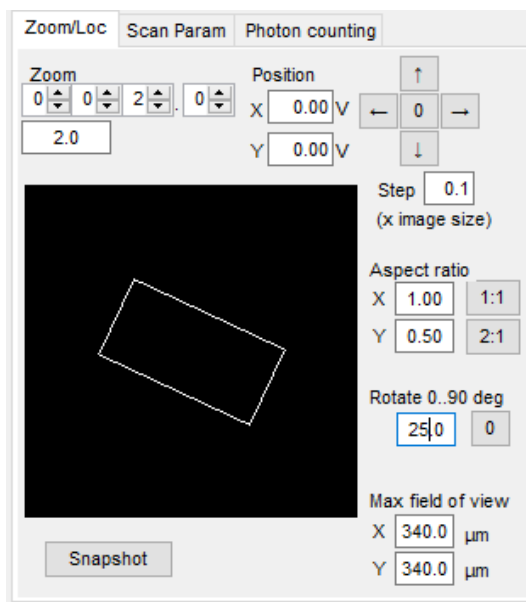
Zoom/Loc tab

This panel controls the scan area. If you set Zoom = 1, the microscope scans the largest area. The value will be anti-proportional to the scan length (Zoom = 3 means it scans 1/3 length). The relative location of scanning is displayed in the panel.

Aspect ratio, rotation, and field of view size can be set here too.

To measure the field of view, image an object, move the object from the top to bottom, while reading the motor position (assume that the motor is calibrated).

Pressing “Snapshot” button will acquire an image with zoom 1 and display it in the panel.



Zoom/Loc Scan Param Photon counting

Zoom

Position X Y

Step (x image size)

Aspect ratio X Y

Rotate 0..90 deg

Max field of view X Y

Scan Param tab

The tab controls various parameters of laser scanning. Line time should be larger than ~1 ms for regular scanning. Number of lines per frame and number of pixels per line can be set in this panel.

In addition, scan voltage range, scan fraction, fill fraction and scan delay of the scanning for the fast scanning axis can be set (**Figure 1**). Scan fraction defined the fraction of forward scanning period. Fill fraction is the fraction that image acquisition occurs. You should use a range in which the mirror motion is relatively linear. The scan delay is the delay of actual mirror motion from the applied voltage. Typically, one can use 0.2 ms for scan delay, ~0.9 for scan fraction, and ~0.8 for fill fraction. If image is significantly deformed, “advanced” check box should be checked and tweak these values.

Zoom/Loc		Scan Param		Photon counting	
32 x 32		256 x 256			
64 x 64		512 x 512			
128 x 128		1024 x 1024			
☐ Advanced setting					
Pixels/Line	128	Voltage range X	5.0	V	
Lines/Frame	64	Voltage range Y	5.0	V	
Line time	1 ms	Fill fraction	0.700		
Pixel time	5.469 μ s	Scan fraction	0.900		
		Scan Delay	0.200	ms	

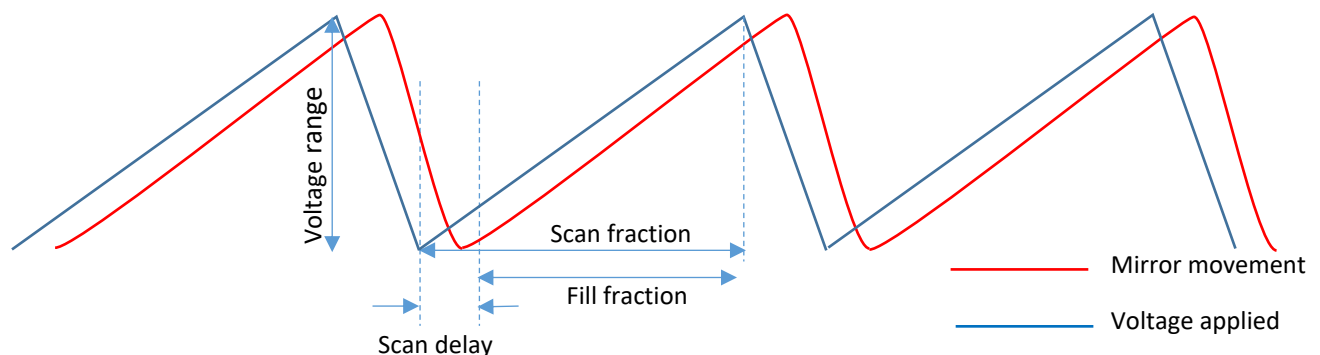


Figure 1: Schematic illustration of command voltage (blue) and actual mirror motion (red).

Resonant scanning

From Version 4, there is an option to use resonant scanning. From version 4.0.23, the phase delay of the resonant scanner can be saved for each zoom (Press the “Save Delay” button. Need to check “Advanced Setting”). For now, “Scan Delay (ns)” needs to be set manually for each zoom. When there is no saved data, FLIMage will try to interpolate the existing data.

“Focus average” will be helpful for resonant scanning.

Bi-directional scan setting is automatic on in resonant scanner. Turning on “Bidirectional scan Y” may help for resonant scanning.

Photon counting tab

This set the parameters for TCSPC boards (up to two boards).

Sync threshold: it sets the threshold for laser pulse detection. It should be set to a value between -50 and -200 mV for typical photo-diode output from a laser company. TCSPC boards typically require negative pulses. If necessary, put pulse inverter. Change the value until the sync pulse rate becomes robust against small shifts in the threshold, and it indicates a value similar the laser pulse rate (~80 MHz for typical Ti:Sa laser).

Channel1 and Channel2 thresholds are for photo-detection. It is typically -2 to -6 mV for un-amplified photo-multiplier tubes. Again, you need negative voltages for these inputs. The value should be modified until the photon count is stable against small shifts in the threshold. Also, if you observe a wavy pattern in the fluorescence lifetime curve, changing the threshold sometimes solves the issue.

The screenshot shows the 'Photon counting' tab in a software interface. It contains four main sections for configuring channels and sync parameters:

- Sync:** Thresh: -50.0 mV, ZC: 0.00 mV, Offset: 7000 ps.
- Sync 2:** Thresh: -50.0 mV, ZC: 0.00 mV, Offset: 7000 ps.
- Channel1:** Thresh: -2.00 mV, ZC: 0.00 mV, Offset: 0 ps, Resolution: 100.0 ps.
- Channel 2:** Thresh: -2.00 mV, ZC: 0.00 mV, Offset: 0 ps, Resolution: 100.0 ps.

At the bottom, there are additional settings:

- Resolution:** A dropdown menu showing '2 (100 ps)'.
- #Time points:** A text input field with the value '50'.
- Mode:** A dropdown menu showing 'T3 (> 40 MHz)'.

ZC-level sets the values for zero-crossing voltage for peak-detection. It can be usually zero, but sometimes improving the quality of fluorescence lifetime curve by putting non zero values (need to be negative value for PicoQuant TH260 Series).

For Pico-Quant cards, it is possible to shift the fluorescence lifetime curve along the temporal axis.

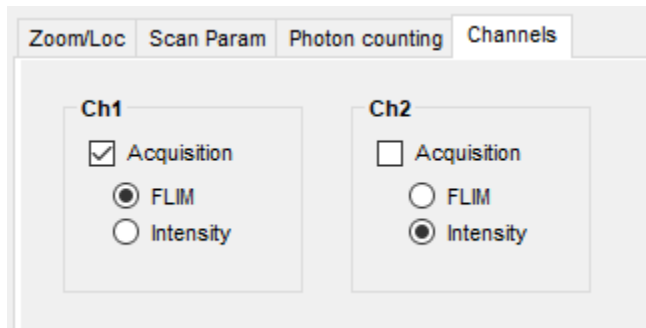
#Time points (only a Pico-Quant card): The number of fluorescence lifetime time points to be acquired.

Resolution: Temporal resolution of fluorescence lifetime acquisition. Typically 100-200 ps for FLIM.

Mode: For a PicoQuant card, T3 mode is used for a high repetition rate laser (> 40 MHz). For a low rate laser (for 3 photon, for example), you may need to use T2. This is an experimental feature for now.

Channels tab

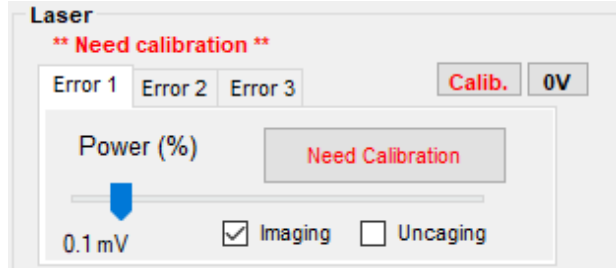
To choose channels to acquire intensity / FLIM, you can use this tab. To save memory, you can disable acquisition for the channels that are not used.



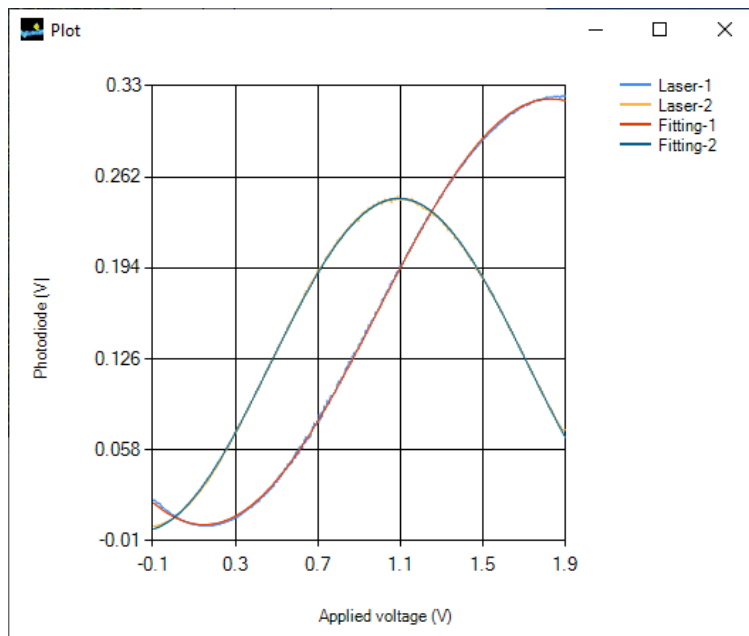
Laser panel

This panel allows you to change the laser intensity by modeling Pockel's cells. The software automatically calibrates it when it starts, but you should calibrate it if necessary (press "calib" button). If the laser is used for imaging, check "imaging" checkbox. If uncaging, check "uncaging" checkbox". For a FRAP experiments etc, you can put both "imaging" and "uncaging".

When calibration is necessary, it will indicate like follows:



This plot shows the calibration result after clicking "calib" button.



Acquisition panel

Focus: Pressing this button will start image acquisition without saving images. It should be used to focus on sample.

Grab: Pressing this button will start image acquisition and save images.

FLIM checkbox: If checked, fluorescence lifetime images will be captured.

Uncaging imaging checkbox: if checked, uncaging will run while image is acquired (only during “Grab”).

Remember setup after each acquisition checkbox: if checked, “Focus” or “Grab” operation will save the current setting.

Analyze after each acquisition: The software will analyze the data after each acquisition.

Laser needs to be pulsed for data acquisition: so, if the software does not detect the correct laser pulse rate, it will warn you as follows.

If you want to start software with an external trigger, check “Ext trigger” checkbox. External port can be set in menubar “Hardware” → “Hardware config”, and External trigger input. The same port need to be connected to the trigger source for all boards (“PFI2” is the default). Internal trigger is on “PFI6” and initiated by “port0/line0”.

If one want to send trigger to external devices, connect port0/line0 to the trigger source.

Saving setup

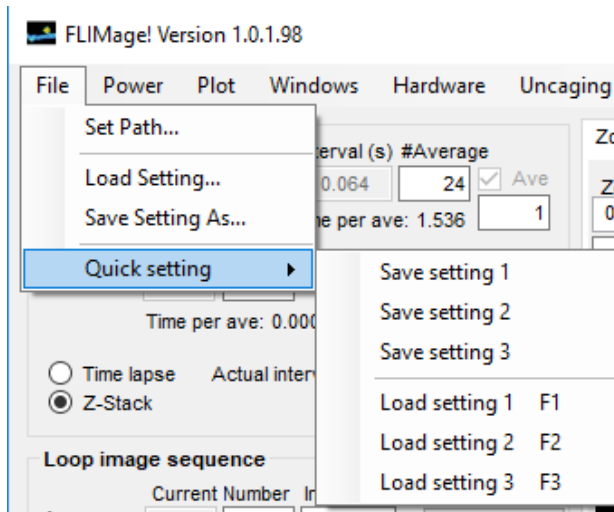
To save current setup, click on “File” menu and then “Save setting as...”.

Loading saved setup

To load saved setup, click on “File” menu and then “Load setting...”.

Quick setting

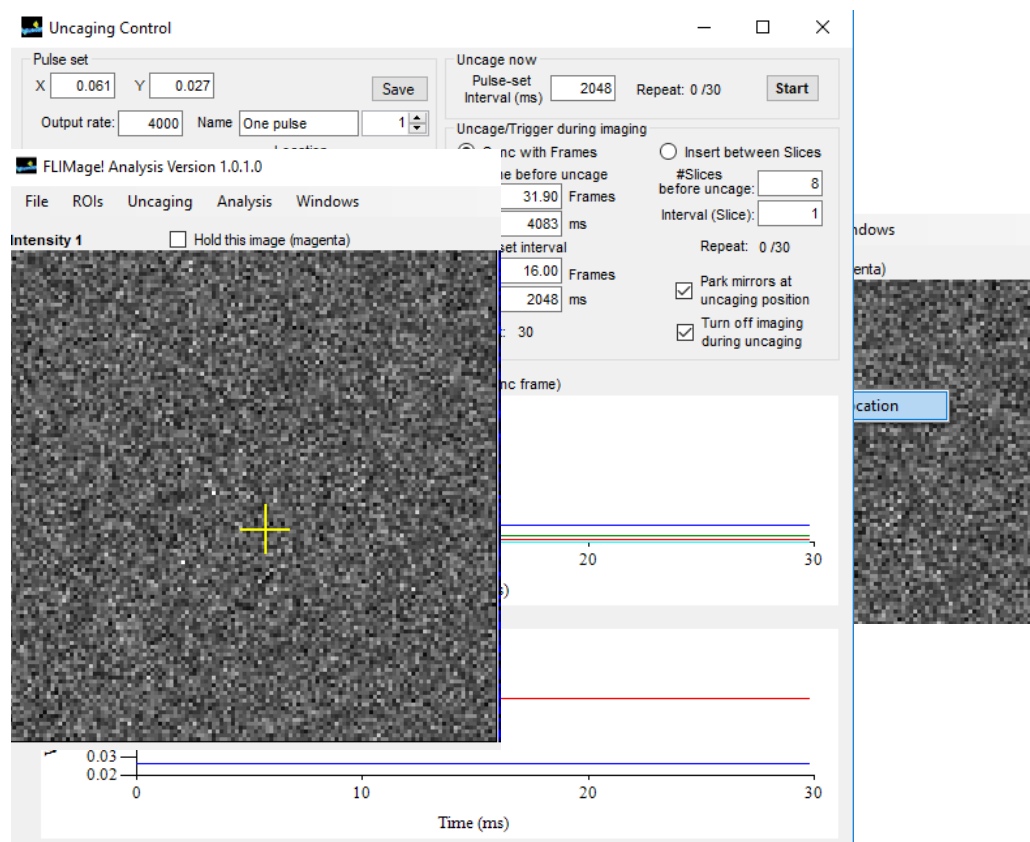
You can save multiple setting 1-3, and quickly change your setting by pressing F1, F2 or F3.



Setting up uncaging protocol

Click on “Tools” menu and then “uncaging control”. Uncaging location will be displayed as “+” on the “image display” window.

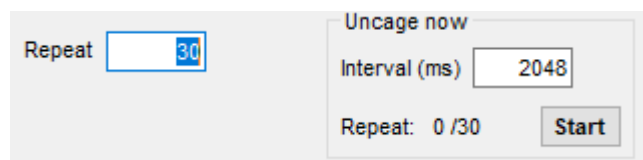
Uncaging panel will appear as follows:



When this panel is open, “image display” window will indicate the location of uncaging with “+” marker. You can change the location by clicking on the image. You can also memorize the position by right click on the “+” and click “Create Uncaging Location” (see below).

In Uncaging Control, X and Y positions indicates the current uncaging location (Yellow cross in the image). You can set pulse numbers, width, power, interval, delay etc. Because shutter can be delayed from actual command, you can set the time shutter opening command occurs (shutter delay).

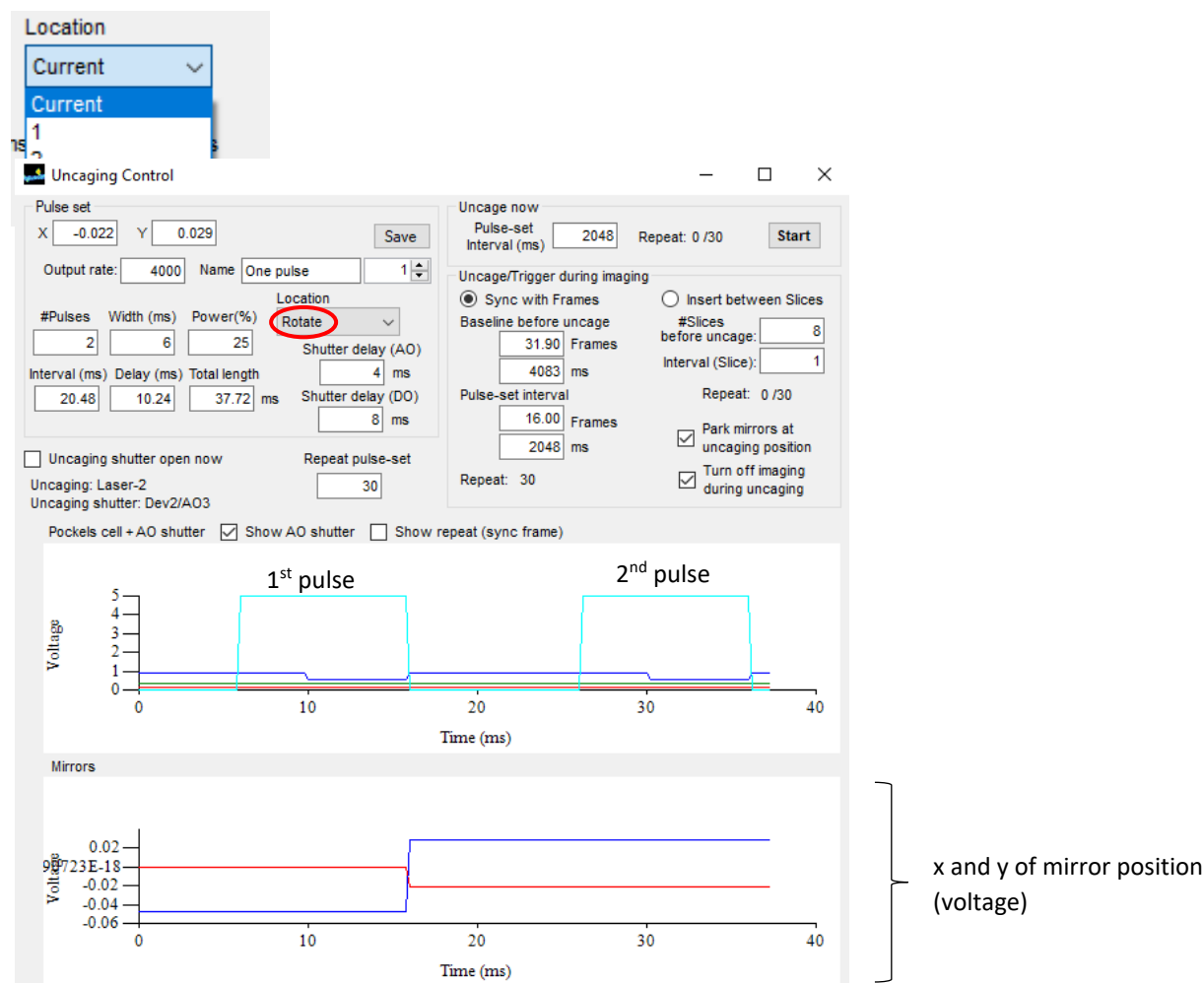
If you just want to uncage, simply press “Start” in the “Uncage now” panel. It will repeat with the interval of the time indicated in the panel. In the following case, it will run the protocol 30 times with 2048 ms intervals.



Digital port can be also used for shutter control. Set this up in “Hardware Control” menu. It can be set up for PMT shutter control (reversed polarity).

Handling multiple uncaging locations

If you have multiple uncaging locations, you can choose which one you want to use.



In addition, "Rotate" means that, if you have multiple pulses, the uncaging location will be changed after every pulse. You can see in the following panel that the mirror position (second graph) changes right after the first pulse to the second position.

Uncaging while imaging

If you want to uncage during imaging, you can check "uncaging while imaging" in the "FLIMage!" panel, and run "Grab". You can either sync with frames or slices (In the time lapse mode).

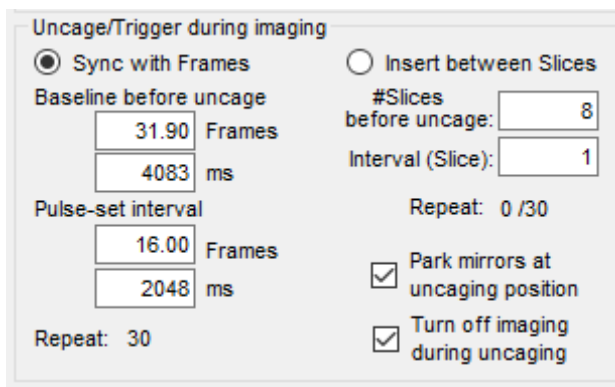
Syncing with slices (former "Page" program)

When syncing with slices, uncaging protocol is inserted between slices. If slices are acquired every 2 sec, the setting in the following panel will take 8 slices (16 s) of background and then repeat 30 times with the interval of 2 sec (1 slice). In this case, these numbers need to be integer.

Syncing with frames

You can also sync uncaging with frames. In this case, scanning will stop during the uncaging and uncaging protocol will run. In this case, it will run together with uncaging for every acquisition if “uncaging while imaging” is checked. The time indicated in “Sync with Frames” panel (see below) is in the unit of “Frame” or “ms” (For example, if 256 ms / frame, the time is divided by 256).

It will repeat the pulse set with the indicated frame / ms. Check “show repeat” checkbox to see the repeated version of pulses.



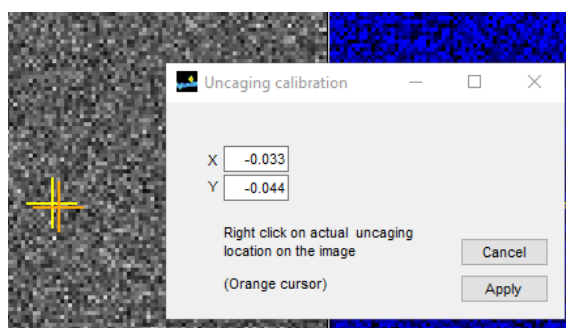
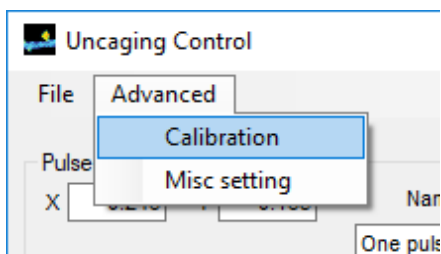
Calibration of uncaging location (in Uncaging control panel)

The uncaging laser and imaging laser should be aligned, but to adjust the difference between the location of these lasers, you can use uncaging calibration panel. This can be opened by clicking on “uncaging” menu and then “Uncaging position calibration”.

One position calibration

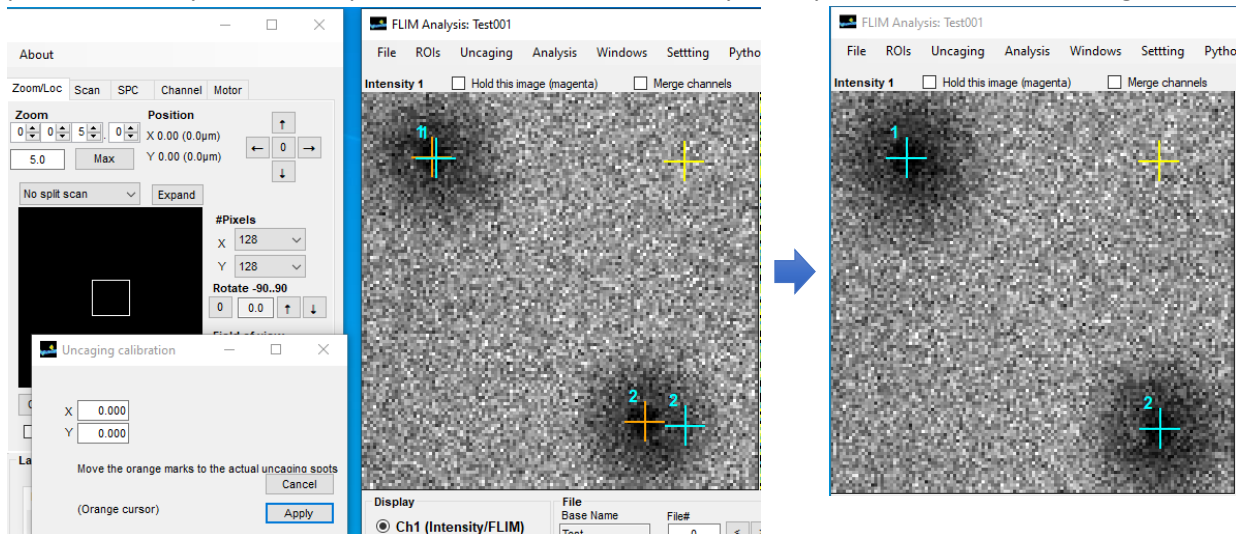
You can image a surface of fluorescence plate and bleach a spot with a single “+” location . If the location and the beached spot is not matched, you need to calibrate the location.

To do so, you can bleach the plate first at a location (yellow +), and then right-click on the center of the bleaching spot (indicated in orange). Repeat until the location of “+” matches to the bleaching spot. Then, press “Apply” button. If the values are larger than 0.1V, you may want to manually put zero and redo this process again.



Two position calibration (ver. 4.0.3)

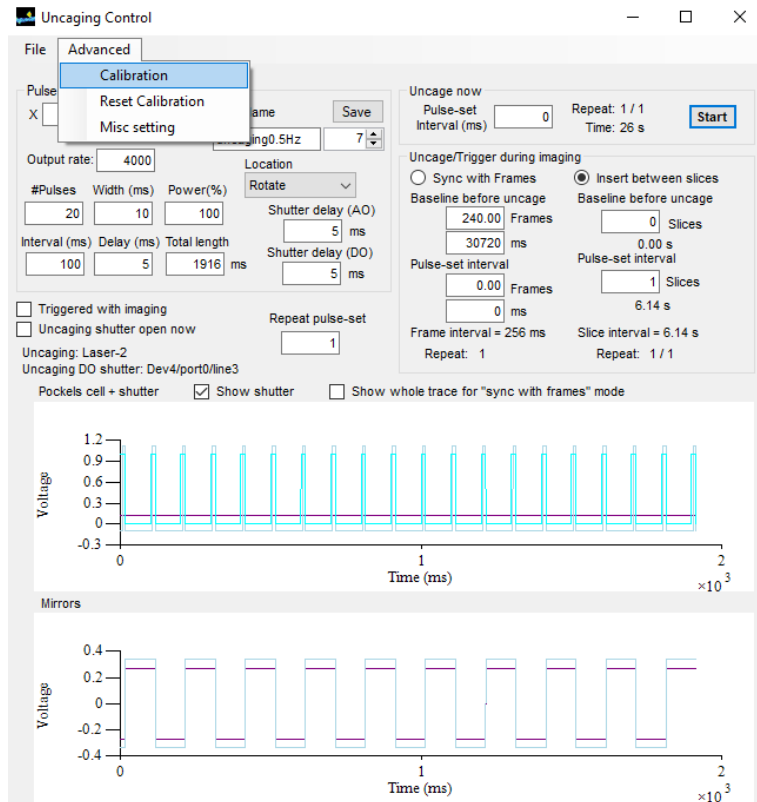
The purpose of this calibration is to determine the translation and magnification of the calibration parameters. If you burn the plate at two or more locations, you may see the difference in magnification



in addition to the translation. In such a case, it is necessary to calibrate both parameters using two locations. As shown in the following screenshot, you burn the plate at two positions placed diagonally apart.

When opening “Uncaging position calibration”, you can see orange cursors, which can be dragged to the actual uncaging positions. Then press “Apply” button. Now you can perform uncaging at the correct positions regardless of zoom size and zoom position.

This is the uncaging condition when performed the two position calibration. Two locations are burned with “rotate”.



Hardware config (Hardware → Config)

This facilitates the setting of hardware. It should be self-explanatory. Don't touch this unless you understand the system.

Hardware config

Scanning (AO)

Board
Dev4

X
AO0

Y
AO1

Scan offset (AO)
☐ Enable

Board
Dev4

X
AO5

Y
AO6

Line clock and trigger for AO

Line clock output
ctr0

Trigger input
PF6

Pockels Cell (AO)

Board
Dev2

Ch 1
AO0

Ch 2
AO1

Ch 3
AO2

Ch 4
AO4

Photodiode input (AI)

Board
Dev2

Ch 1
AI0

Ch 2
AI1

Ch 3
AI2

Ch 4
AI3

N of Channels
2

Sample clock port
PF14

Ext Trigger Input (DI)

Board: all board

Channel
PF12

Marker Input

Start trigger output (DO)

Board
Dev4

Channel
line0

Main shutter (DO)

Board
Dev4

Channel
line1

☐ Open main shutter during calibration

Uncage Shutter Output

Analog

Board
Dev2 (Same as pockels cell board)

Channel
AO3

☐ Enable

Digital

Board
Dev4

Channel
line3

☒ Enable

☐ Active Low

Motor COM port

☒ Use motor

COM17

☐ Sutter MP285

☐ Sutter MP285A

☐ Thorlab MCM3000

☐ Thorlab MCM5000 (no BScope)

☐ BScope

FLIM
☒ FLIM on

Computer ID
4083

FLIM ID
4852

☐ Becker Hicl

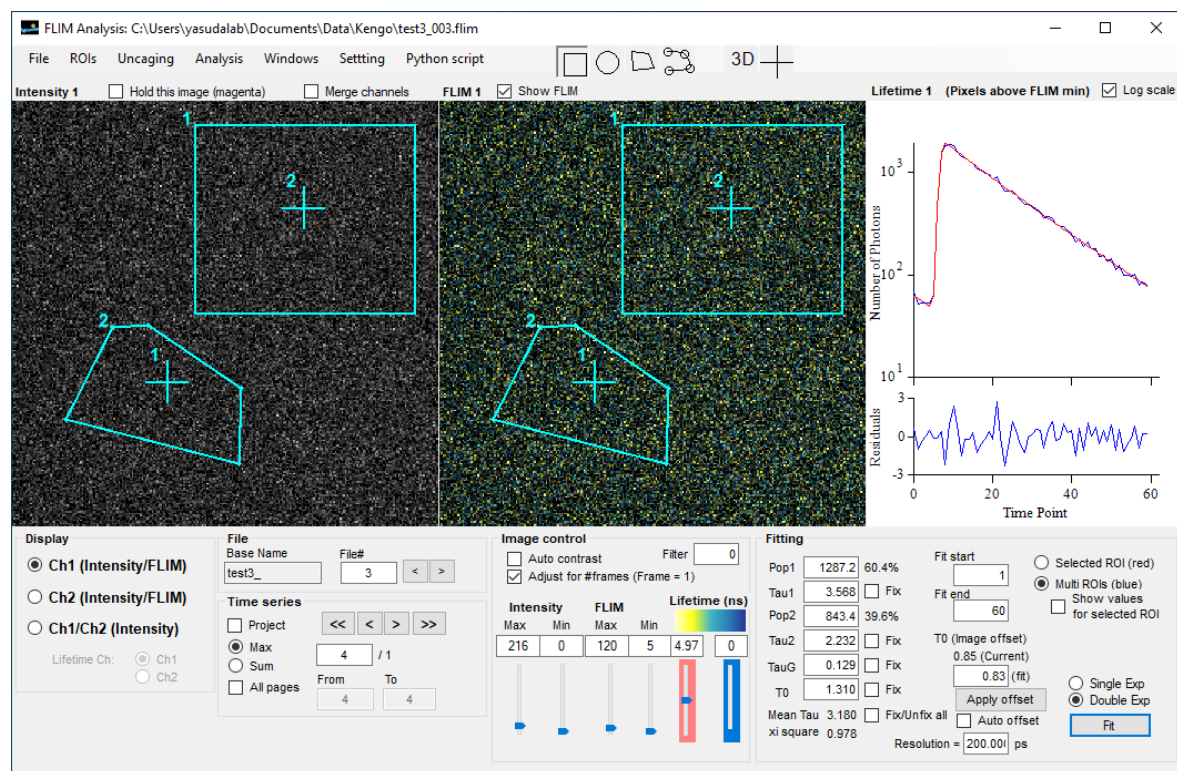
☒ Pico Quant Time-Harp

☐ Pico Quant Multi-Harp

Apply and Restart

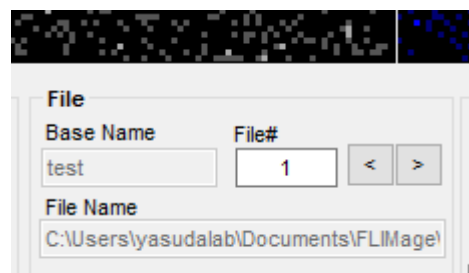
Image display window

This window is to display and analyze acquired images.



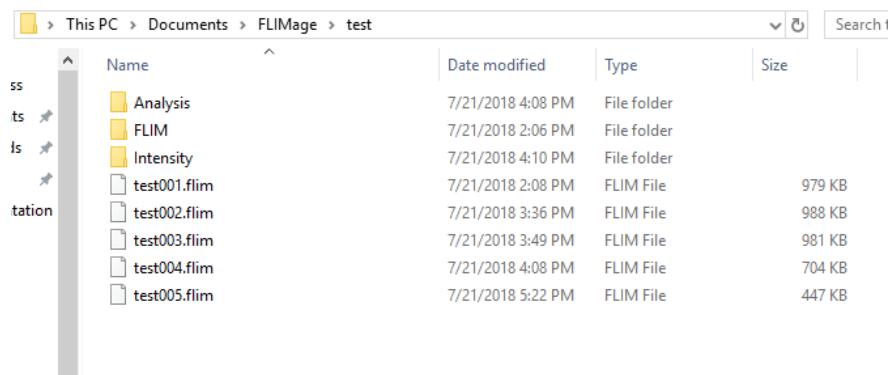
Opening file

Click on "File" menu → "Open FLIM image" (or control – O). If the base name and the folder path is the same, you can open the previous or next image by clicking "<" or ">" buttons on the "file" panel under the images, respectively (see below). For a large file, you may want to stop opening by clicking "Stop opening" button on the "File" panel.



Folder path for analyzed data

All analyzed data will be save in the current folder, the folder in which FLIM images are saved. The software creates subfolders called “Intensity”, “FLIM” and “Analysis” when necessary. All analyzed data are saved in Comma-separated values format, and can be read via Excel and most of other software.



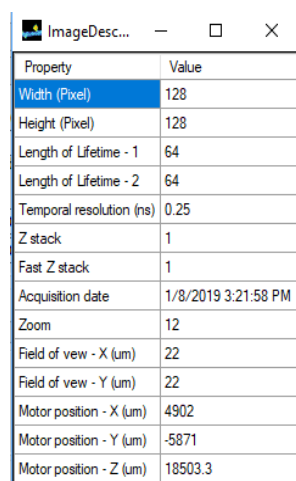
Export tiff images

Click on “File” menu → “Export formatted images (TIF)”. This exports intensity and FLIM images on the display in folders “Intensity” and “FLIM” in the current folder path. Different channels are saved in different files (_ch1 and _ch2), if you check “Channels in separted files” checkbox (right bottom of “FLIMage analysis” window).

If you click on “File” → “Batch processing (Export images & analysis)”, it will analyze and export processed images for the files under the same base name (XXX001, XXX002 etc).

Viewing image setting

Click on “File” menu → “Show image description” to see image information (see below). If you want more detailed list, press control (maybe not readable).



Property	Value
Width (Pixel)	128
Height (Pixel)	128
Length of Lifetime - 1	64
Length of Lifetime - 2	64
Temporal resolution (ns)	0.25
Z stack	1
Fast Z stack	1
Acquisition date	1/8/2019 3:21:58 PM
Zoom	12
Field of vew - X (um)	22
Field of vew - Y (um)	22
Motor position - X (um)	4902
Motor position - Y (um)	-5871
Motor position - Z (um)	18503.3

Display uncaging location On/Off

Click on “Uncaging” menu → “Show uncaging position”.

Max or sum projection

You can check “projection” checkbox in the “Frames / Slices” panel to display projection.

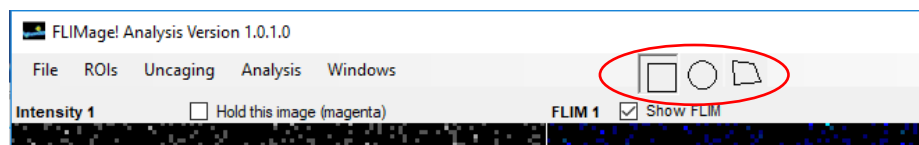


Frames/Slices navigation

You can navigate slices or frames by clicking “<” or “>” in “Frames / Slices” panel (see above). “<<” and “>>” is moving 10 frames/slices.

Region of interest

Click on the image and drag to the left bottom. You can draw a region of interest (ROI). You can choose to create a shape of either rectangular, circular or polygonal. Use the toolbox above the images (red circle in the following figure).



The created ROI is called “selected ROI” and displayed in red. If you right-click onto the ROI, you can create one of “multi ROIs”, which is indicated in cyan.

Double clicking on the window will delete the “selected ROI”. To delete a multi-ROI, right click on the ROI you want to delete, and follow the instruction.

To delete all multi-ROIs, right-click on the window outside of ROIs and choose “Remove All Rois” in the pulldown menu.

Saving ROIs

To save multi-ROIs, click “ROIs” menu and then click “Save ROIs”. This will save ROIs in {Current path}/Analysis. To recover multi-ROIs, click “Recover ROIs” in the “ROIs” menu. Saving/Reading ImageJ style ROIs is possible (in ROIOs menu).

Fluorescence lifetime curve and fitting

Fluorescence lifetime curve is displayed in the right panel. The curve is displayed from red ROI (selected ROI). If you choose to show multi ROIs, it will display all pixels in the multi ROIs. You can choose to select individual multi ROI.

It should be noted that black pixels (pixels with intensity values under the lower threshold) will be NOT included in the curve. This is because low-photon pixels often include little information but a lot of noise.

Fitting will be performed either on selected ROI or all multi ROIs (you can select with radio buttons on the “Fitting” panel).

“Fit start” and “Fit end” defines the fitting range (in time point). Also, the lifetime window will display only this range.

Fitting will be performed with Gaussian instrument response function (width = tauG) convoluted with exponential curve. You can choose to fit with double- or mono-exponential curves (decay constants = tau1 and tau2, populations pop1 and po2). Time offset (T0) will be also fitted.

Fitting will be performed when you click “Fit” button. At the same time, it will save all parameters in “Analysis” folder in the current folder path.

You can fix any of these parameters.

The fitting function is defined as follows.

For single exponential curve:

```
y1 = pop1 * Exp(tauG^2 / tau1^2 / 2 - (x - t0) / tau1);
y2 = Erfc((tauG^2 / tau1 - (x - t0)) / (Sqrt(2) * tauG));
exp_gauss(x) = y1 * y2 / 2;
```

We usually add fluorescence lifetime decay from 1 pulse before:

```
exp_gaus_corrected (x) = exp_gaus(x) + exp_gaus(x-IPI),
```

where IPI is the iter-pulse interval (for 80 MHz, it is ~12.5 ns).

For double exponential curve:

```
y = pop1 * exp_gauss_corrected (x, tau1) + pop2 * exp_gauss_corrected (x, tau2)
```

FLIM colormap image

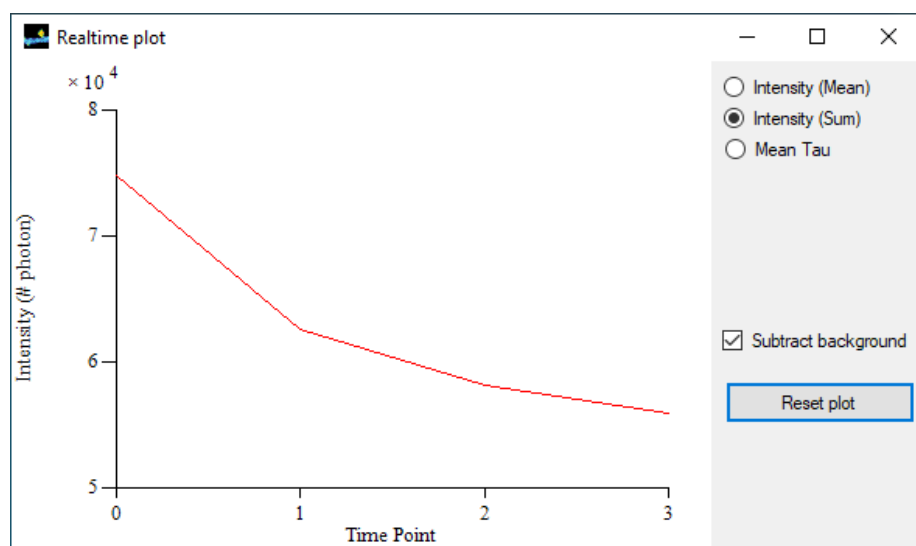
To correctly display colormap, you need to fit the fluorescence lifetime curve and then apply time offset (T0) by clicking “apply” button.

Correcting drift for images

Analysis → Align pages will align image pages within the same image. For series of images (like XXX001.flim, XXX002.flim etc), use Analysis → Make Movie from Series. This will make Z-projection (Max or Sum) for each image (if Z-stack) and align. For T-stack or any time series, it will handle each page as a single image.

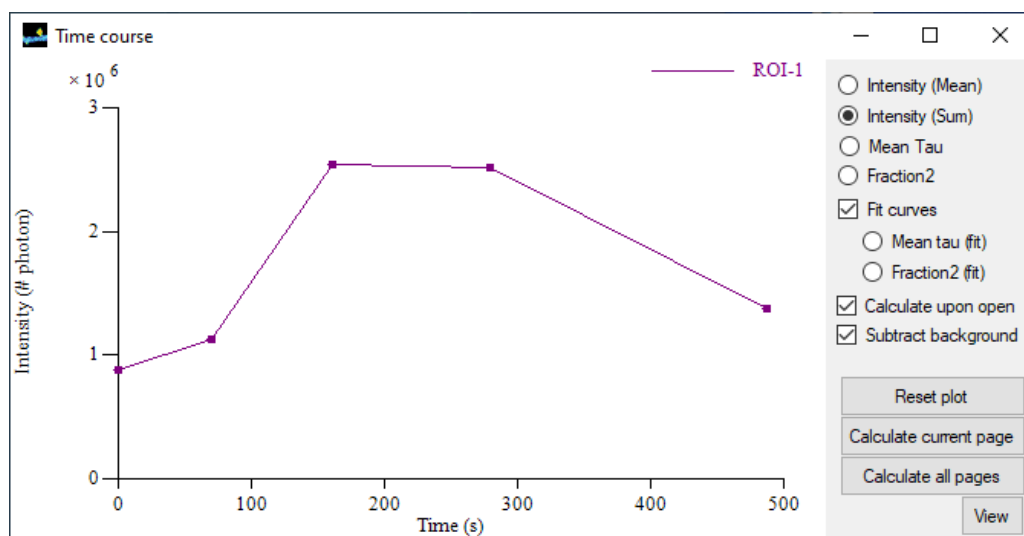
To save the collected and corrected file, do File → Save FLIM Image. For ImageJ readable image stack, use File → export formatted images.

Realtime plot window



Time course window

This window is to get time course of the time-lapsed images.



Intensity value INCLUDES black pixels. It is sum of fluorescence in the ROI.

Mean tau is defined as the average photon arrival time subtracted with time offset (T0). Black pixels (pixels with intensity values under the lower threshold) are NOT included in the curve. From the lifetime curve (time vs fluorescence) in each ROIs, it is calculated as:

$$\text{Mean tau} = \sum_i (\text{time} * \text{fluorescence}) / \sum_i (\text{fluorescence}) - t_0,$$

where i is all pixels in the ROI and above the threshold.

Fraction2 is the fraction of tau2 population:

$$\text{Fraction2} = \text{pop2} / (\text{pop1} + \text{pop2}).$$

Its approximate value can be obtained from mean tau (tau_m0) as:

$$\text{Fraction2} = \text{tau1} * (\text{tau1} - \text{tau_m0}) / (\text{tau1} - \text{tau2}) / (\text{tau1} + \text{tau2} - \text{tau_m0}).$$

If you click “Fit curves”, the software will fit fluorescence lifetime curves in each image. Values obtained by fitting will be stored as “Mean tau (fit)” and “Fraction2 (fit)”. Mean tau in this case is defined as:

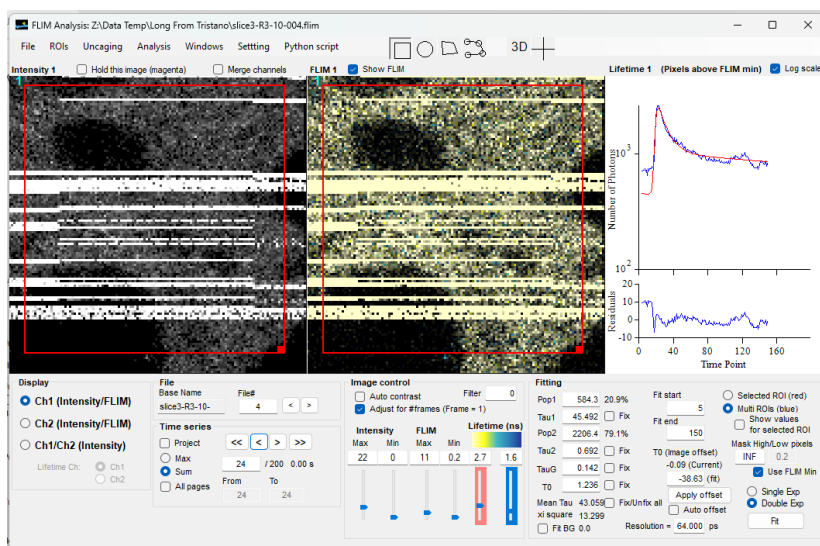
$$\text{Mean tau} = (\text{tau1}^2 * \text{pop1} + \text{tau2}^2 * \text{pop2}) / (\text{tau1} * \text{pop1} + \text{tau2} * \text{pop2}).$$

All values are automatically saved in “Analysis” folder in Excel (csv) format.

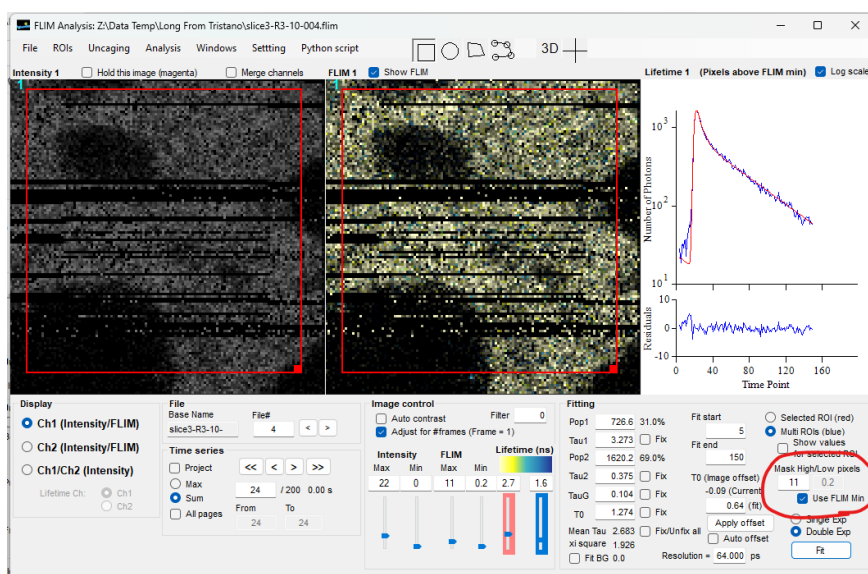
Masking high/low intensity pixels (After 4.0.29)

Low-intensity pixels are usually masked for lifetime calculation to avoid noise. The threshold is determined by “Min” in FLIM intensity panel.

On rare occasions, single-photon counting can cause significant noise (see below) with high intensities. In these situations, one can remove these pixels by setting a higher threshold.



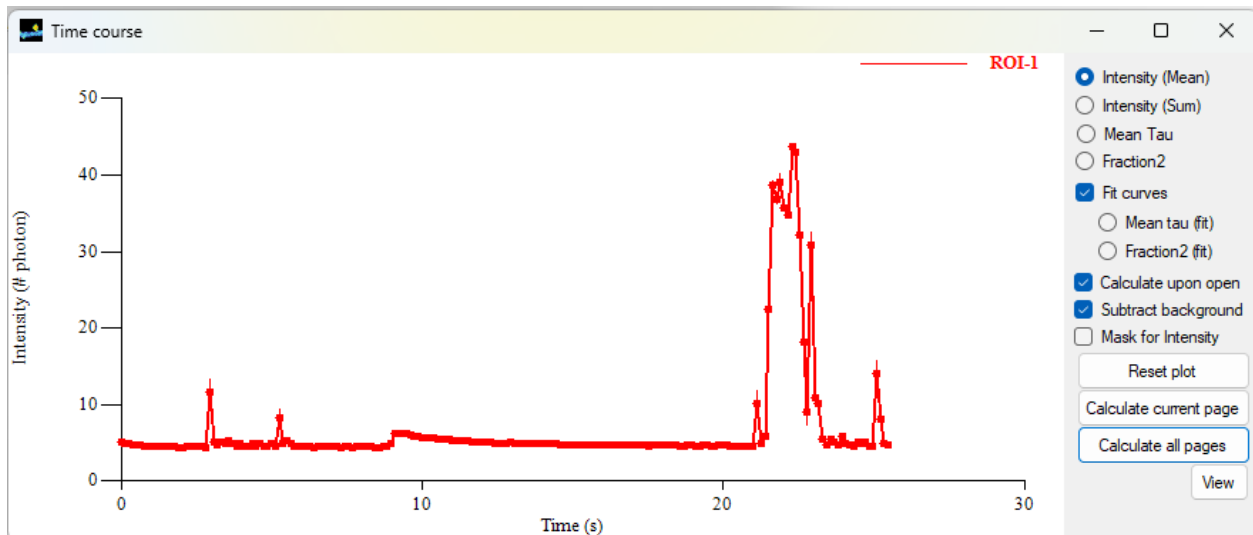
The mask can be set by setting a high threshold for masking (red circle). In addition, one can put a low threshold independently of the image threshold by checking off “Use FLIM Min” and entering a number (> 0). Negative numbers will be ignored for a high threshold.



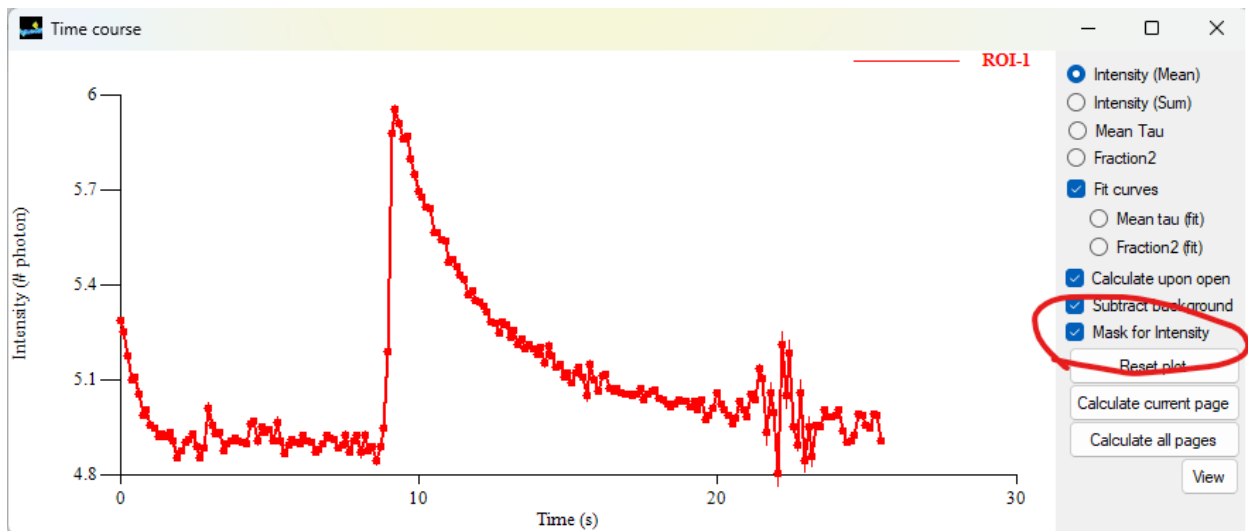
This setting can significantly improve fitting.

Additionally, if you want to use this mask for intensity calculation in the time course plot, you can check “Mask for Intensity” in the Time Course window.

(before check)



(after check)



Scan calibration menu for resonant/bidirectional scanning
(Tools → Scan calibration) (After version 4.0.32)

These tools automatically adjust **Scan Delay** to align **even vs odd lines** in **bidirectional X scanning** (galvo bidirectional, sine-wave scan) and **resonant/polygon scanning**. They are intended for **stationary calibration targets** (beads, grid).

Quick auto scan delay (even/odd)

Use when: you are already “close” and want a fast touch-up (small correction). **What it does:**

- Measures the **even/odd X shift** from the currently displayed intensity image (small search around 0).
- Converts shift → delay correction and updates **Scan Delay**.
- If focusing is running, it restarts focus once to apply the new delay.

How to use:

- Start **FOCUS** on beads/grid (recommended).
- Set **FocusAverage** high enough for stable signal (typical: 20–100 for resonant).
- Select the channel with best contrast.
- Run: Tools → Scan calibration → Quick auto scan delay (even/odd)
- If the status message warns “shift near limit”, use the full routine below.

Auto scan delay (even/odd)...

Use when: the mismatch is large, alignment is confusing by eye, or reconstruction is degraded (e.g., strong blanking/warping / photons dropped). **What it does:**

- Runs a **coarse → fine scan** over possible Scan Delay values.
- For each candidate, it briefly **restarts focus**, acquires an averaged frame, and scores **even/odd correlation**.
- Chooses the delay with the best score and applies it (with an optional final fine adjustment).

How to use:

- Start **FOCUS** on beads/grid.
- Increase **FocusAverage** until the image/score is stable.
- Run: Tools → Scan calibration → Auto scan delay (even/odd)...
- Wait for completion; progress is shown in the status text.

Notes / troubleshooting

- **Image display required:** if you see “Image display is not available yet”, open the image window (Windows → FLIM_analysis) or start focusing first.
- **If the score is unstable / fails:** increase averaging, use brighter beads/grid, avoid saturation, and ensure there is enough structure/contrast.
- **Where to see the result:** the **Scan Delay** field updates (displayed in ns). Internally it updates the appropriate parameter depending on scan mode (resonant/polygon vs galvo).

Line scan trace mode (polygon-based)

Overview

Line scan trace lets you scan along a **closed polygon path** (the polygon edges) at approximately **constant speed** (longer edges take proportionally more time). Data is saved using the existing .flim format. The number of lines captured will be (Number of Frames) x (Number of lines per frame), and saves every frame.

- **Focus:** always normal **2D imaging** (for drawing/visualizing the polygon).
- **Grab:** runs the **line scan trace** acquisition when enabled.

Requirements / limitations

- **Regular galvo–galvo only**
- **Not supported** with **resonant scanning** or **polygon scanning** enabled.
- The scan path is a **closed loop** (no flyback segment).
- **No scan delay** is used for the trace mode.
- **EOM is always ON** during the trace (no XY shading/blinking along the path).

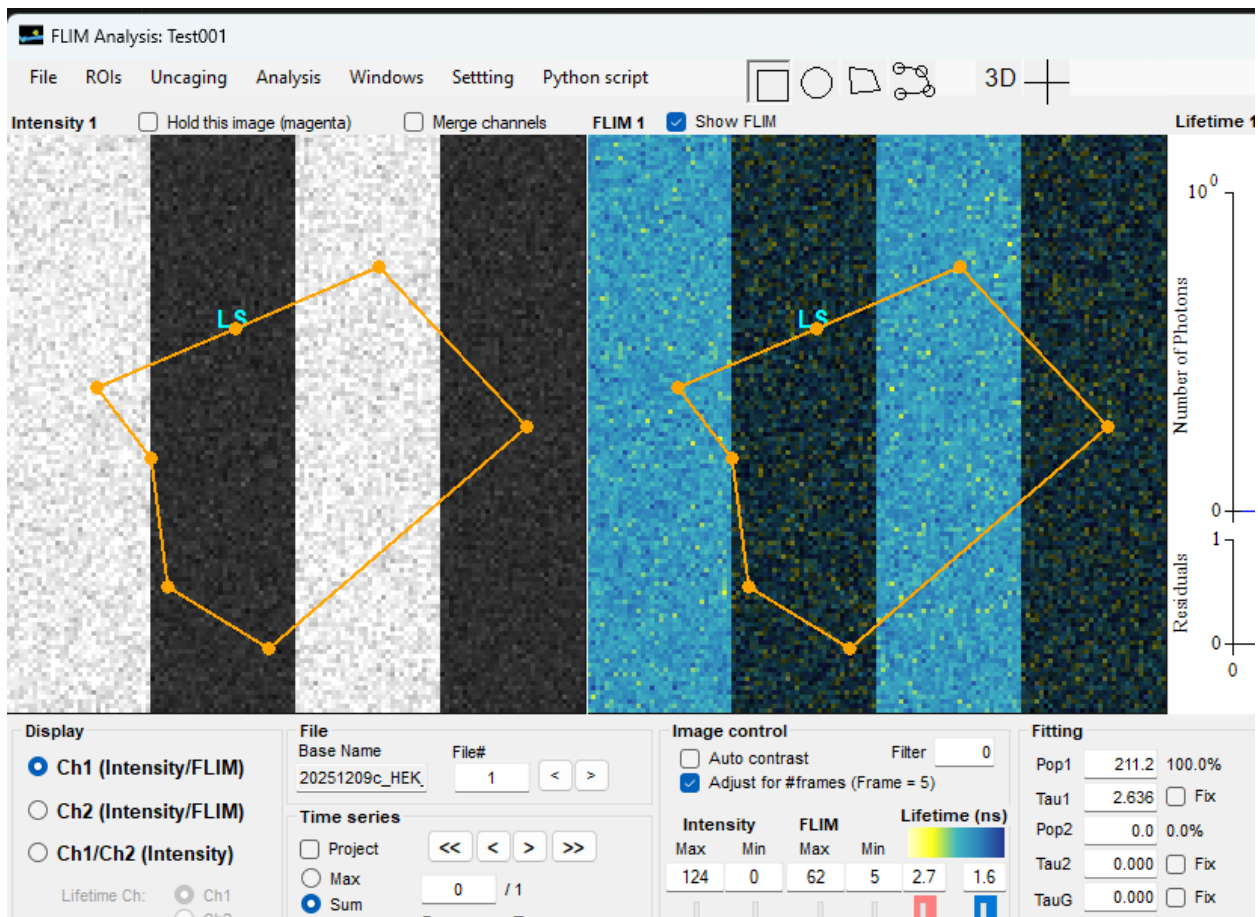
Creating the line scan trace

1. Click **Focus** to show the 2D image.
2. Draw a **Polygon ROI** in the image window. (Ellipsoid and rectangular shapes also work, but will be converted to polygon).
3. **Right-click the polygon ROI** → choose **“Use this trace for line scanning”**.
 - The trace is shown with the **“LS”** label.
 - **Unselected** LS trace appears **orange**.

- **Selected** LS trace appears **red** (still labeled “LS”).
- The LS trace is stored separately and **does not appear in the normal MultiROI list** (no ROI numbering / timecourse confusion).

To remove it:

- **Right-click the LS ROI → “Remove this line scan trace”.**



Editing the trace

- Click the LS ROI (it becomes **red** with “LS”).
- Drag vertices/edges as usual.
- On mouse release, the stored line-scan vertices are updated automatically; the **next Grab** will use the updated trace.

Running acquisition

1. In the main window, enable “**Line scan trace**” (LineScan_CB).
2. Ensure **resonant scanning** and **polygon scanning** are **OFF**.
3. Click **Grab**.
 - Acquisition is run as a **1-line frame** while scanning along the closed polygon edges at constant speed.

Saved header fields (for reconstruction)

The minimal polygon vertices (not the full voltage waveform) are stored in the .flim header:

- State.Acq.LineScanArrayX (double[])
- State.Acq.LineScanArrayY (double[])

These are **image pixel coordinates** of the polygon vertices; the path is interpreted as **closed** (last vertex connects back to the first).

Fast saving mode

Since FLIM data is extremely sparse, we usually compress the data before saving. However, for fast scanning, it may be ideal to save in much faster mode. In that occasion, you can check “Fast Saving” check box under the “Channel” tab. This will speed up the saving process by x20 times but also the file size can be orders of magnitude bigger (can be ~100 times).

FiberPhotometry mode (single-point TCSPC) — Setup & Usage

FiberPhotometry mode acquires **single-point data** (1×1 pixel at x=0, y=0) by **binning photons in macro-time**. Each macro-time bin becomes one “frame/page” in a .flim file. Key points:

No image rendering (prevents bitmap/stride errors). Only the **lifetime decay curve** + **realtime plot** are updated.

Data is saved as a standard **.flim TIFF stack** (1×1 pixels, N pages) so the normal analysis pipeline works.

CSV is **not** written during acquisition; use **Export → FiberPhotometry CSV...** after acquisition.

1) Enable FiberPhotometry mode

FiberPhotometry is enabled when the microscope system is set to FiberPhotometry.

Init file / header setting:

```
State.Init.MicroscopeSystem = "FiberPhotometry"
```

When enabled, FLIMage forces acquisition geometry to **1×1** in the decode/saving path.

2) Set sampling interval (bin interval)

The bin interval is stored as:

```
State.Acq.fiberBin_ms (milliseconds)
```

In the main window:

The **FrameInterval** box becomes **editable** only in FiberPhotometry mode

It displays **fiberBin_ms / 1000** (seconds)

How to change:

Click **FrameInterval**

Type the desired interval in **seconds**

Press **Enter**

This updates:

```
State.Acq.fiberBin_ms = FrameInterval_s * 1000
```

3) Start trigger mode (Soft vs External)

Start mode is selected via the existing external trigger setting:

Soft trigger (start on first photon):

```
State.Acq.externalTrigger = false
```

```
State.Acq.fiberStartMode = 0
```

External trigger (start on marker):

```
State.Acq.externalTrigger = true
```

```
State.Acq.fiberStartMode = 1
```

These values are saved in the .flim header (ScanParameters fields), so you can read them later.

4) Set number of time points

State.Acq.nFrames is treated as the number of FiberPhotometry bins/pages to acquire.

5) What you should see during acquisition

Displays:

Lifetime decay curve updates for the current channel.

Realtime plot shows the selected metric vs **Time Point** (index).

Intensity (Mean)

Intensity (Sum)

Mean Tau

Not displayed:

Intensity/FLIM images are not drawn in FiberPhotometry mode.

6) Saving behavior

.flim output

FiberPhotometry Grab writes a standard .flim file:

1×1 pixel

multiple pages (one page per bin)

header includes FiberPhotometry fields:

State.Acq.fiberPhotometryMode

State.Acq.fiberBin_ms

State.Acq.fiberStartMode

Timestamps:

Each page's acquired time advances by **fiberBin_ms** (so offline time axes are correct).

File splitting (important)

Normal imaging can split acquisition into multiple files using:

State.Acq.maxNFramePerFile

In FiberPhotometry mode, FLIMage forces **one file per grab** by overriding rollover behavior so:

filename does **not** change every N frames

fileCounter does **not** increment mid-acquisition

7) Export FiberPhotometry as CSV (after acquisition)

CSV export is done after the .flim is saved:

Open the saved .flim file in the Image Display window

Use:

File → Export → FiberPhotometry CSV...

Choose the CSV filename/location

Exported CSV columns:

MacroTimeISO

Elapsed_s

Channel

MeanIntensity

SumIntensity

MeanLifetime_ns

Full decay curve columns: ps_0, ps_100, ... (microtime axis in picoseconds)

8) Simulation mode (PicoQuant SIM_PQ)

Turn on simulation

```
State.Init.FLIM_mode = "SimPQ"
```

What is simulated in FiberPhotometry + SimPQ

Simulator emits photons for a **single point** (1×1)

Intensity is sine-modulated:

$$\text{inten} = \text{base} + \text{amp} * \sin(2\pi f t + \text{phase})$$

Default sine frequency:

1.0 Hz (fiberSineFreq_Hz = 1.0)

Where it is configured:

Hardware\TCSPC_Controller\TCSPC_Decode\TCSPC_Decode.cpp

In the SIM_PQ setup block for fiber mode:

```
simpq->fiberSineFreq_Hz = 1.0;
```

Important:

Sine uses **continuous time across frames** (phase does not reset each bin).

Using python script

In Documents\FLIMage\Python script, there are several python scripts useful for analyzing the data. These scripts run on a standard Anaconda 3 environment.

FLIM_pipeClient.py

This is a python Class to communicate with FLIMage. See RemoteControl_Tutorial.docx for API.

FLIM_PlotScript.py

This is an example script that uses FLIMageFileReader.py to open '*.flim' images created by FLIMage, calculate lifetime and plot.

FLIMageFileReader.py

This is a python class that provides function to read files (like '*.flim' or '*.tif' created by FLIMage! software. It also include function to calculate lifetime, intensity and lifetime map. Detailed parameters are stored in FileReader.State.

This script requires pylibtiff package. Although anaconda includes this package, you need to run 'pip' to setup the environment:

```
>> pip install pylibtiff
```

in a shell programs with python path included (for example, 'anaconda prompt' software can do this).

In addition, pylibtiff package requires LibTiff.dll that is not included in the installer. You may download the source from <https://gitlab.com/libtiff/libtiff> (latest version 4.5.0 as of May 2023) and cmake using Visual Studio or CMake, then put it in python directory.

ReadFLIMageCSV.py

This is a python class that provides functions to read and average all time course files (.csv files) in a directory.

ReadFLIMageCSVGUI.py

This program is a GUI backend for ReadFLIMageCSV program.